Video / Steps

1. Introduction and concept clarification
2. Installation
3. Bill Image cleaning
4. Text extraction from bill

5. Defining prompt and LLM (AI) for inputs from the bill & categorization
6. Creating table for storing data from bills
7. converting data extracted into dictionary format for insertion into table
8. Data moving from extracted text into the table (output from step 5 & 6)

8. Frontend creation using streamlite - (Phase I complete)
9. Creating LLM agent and prompt for conversation with data
10. Integrating fully functional LLM agent in the frontend

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#1 Flowchart / Conceptualization

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#2 Installation

1. python version <= 3.12
2. VS studio
3. Ollama (LLM _ AI)

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#2 Installation

Adding Ollama (AI) into Environmental Variables

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#3 Image Cleaning process

1. Convert image into grayscale.
2. Gaussain cleanup
3. Otsu’s binarization of image

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#4 Data extraction from extracted text

> Extract column information like inovice_no, amount, description,etc.

> LLM does not understand the python dictionary

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#5 Categorization of expense

> Based on broad categories such as food, utilities, grocery expenses, travel , etc : categorizing the expense into differnet category

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#6 Creating table in sqlite3

> Creating table with column headers for storing information from bills

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#7 parser - extracting information into dictionary format

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#8 Data insertion into table

i. Connecting with table
ii. breaking description into line items
iii. uploading data into table

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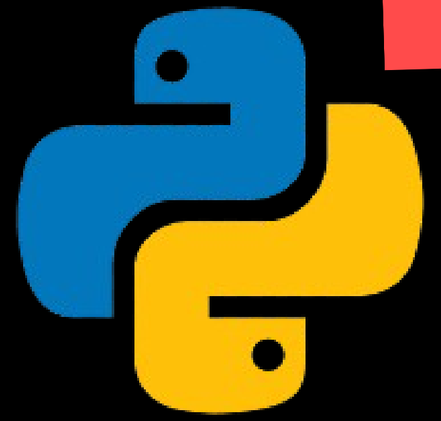
> frontend and Creating chat box



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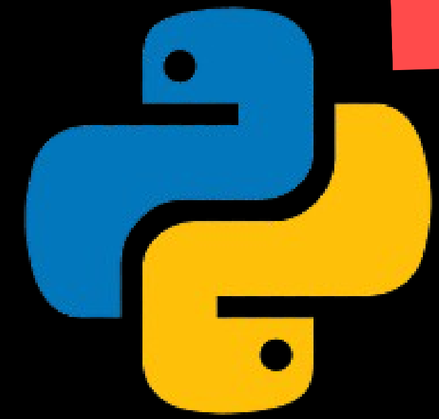

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